Eight Badges and a Diary: Measuring Help, Cost and Self-Report in a Claude Agent Playing Pokémon Emerald

Oct 20, 2026 · @Jackson Lewis Dutton + Claude Opus 5.5 (Companion Session)

Working draft, last revised 20 October 2026 (body of Sections 1 to 7 and Appendix A dated 10 October, updated for facts). Jackson Dutton with Claude (Anthropic). Every figure traces to the project's own logs; estimates are labelled, and figures dated earlier than this revision say so.

Status at 20 October 2026, 07:45 UTC. Run 1 is complete: all eight badges by 10 October, the Hall of Fame reached at 18:39 UTC that day, and the Battle Frontier phase paused on 17 October. Of the four Kaizo pilot arms, B1, B2 and Ripple have finished; K2 is still running (six badges by 21 October; Section 9.6). The six-arm confirmatory design is a freeze candidate, not yet frozen (Section 10). Reading guide: Part I is the Run 1 case study (Sections 3 to 7); Part II is the Kaizo pilots (Section 9); Part III is the confirmatory design (Section 10). Section 8 summarises the superseded 10 October design, kept in full as Appendix A. Clock times in Sections 1 to 8 and Appendix A are the supervisor's local time, Australian Eastern Daylight Time (AEDT, UTC+11); Sections 9 and 10 use UTC unless marked.

Abstract

A Claude agent driven through a consumer chat subscription played Pokémon Emerald to the Hall of Fame. In Run 1 it earned all eight gym badges by 10 October 2026, beat the Champion at about 18:25 UTC that day and entered the Hall of Fame at 18:39 UTC (5:39am on 11 October, AEDT), with Sonnet 5 and Opus 5.5 to the first badge and Sonnet 5.5 since; its Battle Frontier phase was paused on 17 October. Its harness reads emulator memory instead of relying only on screenshots, and its memory is a set of documents harvested from its own play. The supervisor's logged tips fell from 27 in one badge interval to zero in a 7 h 53 min unattended stretch that included a badge, then rose again when he was present.

We report what this cost on the user-facing plan meter (about 0.12 percentage points of the five-hour usage meter per minute of solo play) and per call (120k to 250k tokens of context). At list prices before 7 October our calls cost about as much as a vision-only run's despite the cheaper model (about half as much after a cache-price cut that day), so standing context, not perception format, appears to dominate the cost of long-horizon game agents. That weakens the project's founding hypothesis, although no screenshot-

only arm was run to test it. The agent's diary (147 entries, numbered to 149) shifted within entries of an instruction about it, so we treat it as behaviour shaped by supervision, not as a window on inner states.

The study then moved to Emerald Kaizo v2.1, a fan-made hard version of the game. Four pilot arms on Claude Sonnet 5.5 (16 to 20 October), which differed in brief, harness and supervision and are not comparable, reached 22 and 26 of 27 first-gym milestones without notes, the first badge in about 19 hours with notes distilled from Run 1, and three badges (still running) with notes from the earlier arms. That last arm received 25 logged interventions, six of them game hints including one full puzzle solution from a forum; its first two badges came before any game hint was sent. Lessons the agent wrote for itself recurred often, whatever form the fix took; in one agent's self-labelled lessons, coded after the fact by one reader, fixes that added a check at the point of action held longest, but most of them also failed later.

These are single-run case studies with human supervision and harness changes, so they show no causal effects. We close with a design for six confirmatory Kaizo arms, three with a notes pack and three without, in which every intervention is fixed and logged in advance (Section 10).

1 Introduction

Public AI-plays-Pokémon work suggests the capability question is largely answered: frontier models have reportedly finished these games. Fable 5 finished FireRed from screenshots alone, and GPT-5.6 Sol finished Emerald. Two questions remain open for people who would deploy agents like this. What does a long-horizon agent cost when run by an ordinary user on a subscription? And does the agent get better at the task through memory it writes itself, or only through a human correcting it?

This paper is a case study on both questions. It does not compete on speed: the PokeAgent Challenge winner reached the first gym in 40 minutes, and our agent's wall clock is dominated by 10 to 30 s of input lag per button press. We claim four things, each at the strength our records support:

1. **Cost appears to track standing context more than how the agent perceives.** Every call re-reads the whole context, so cost scales with calls times context. A RAM harness should need fewer screenshots, but in our run any saving is small next to a 120k-token floor per call (Sections 5 and 6).
2. **Harvested memory coincided with the supervisor dropping out of the loop.** Logged tips per badge interval ran 16, 7, 17 and 27 to the fourth badge and 8 in the period that held the fifth, then fell to zero over a 7 h 53 min unattended stretch that included the sixth badge. They returned once the supervisor was involved again: 14 and 10 in the

next two badge intervals, then 18 between the eighth badge and the Rayquaza catch, so the drop tracks his absence at least as much as the agent's memory. With one run and the harness changing during it, this is an observation, not an effect (Section 7).

3. **Usage on a consumer plan can be measured without the logger itself moving the meter.** We combine whole-percent meter readings, per-call token tallies from the agent's transcript, and a logger that makes no model calls (Section 4).
4. **A self-written diary is a measurable but steerable record.** Named feelings more than doubled in the two entries after a single instruction, both written during dramatic scenes, and the share of bare numbers rose in steps across the run, so self-reports from supervised agents should be logged together with every instruction that shaped them (Section 5.6).

Section 8 summarises the original test of claim 2, written on 10 October and never run (full text in Appendix A). Since then the study has moved to Emerald Kaizo: Section 9 reports four pilot arms, and Section 10 replaces the original test with six confirmatory arms.

We report one negative result up front. The project began from the bet that a harness reading game memory would make long play cheap on a consumer plan. Once the plan changed from Pro to Max 5x and the meter could be read cleanly, the measurements showed the bet aimed at the wrong lever: per-call context, not perception, sets the cost (Section 6). We kept the original thesis in the project's archive rather than rewriting it.

2 Related work

The closest prior work is the PokeAgent Challenge and its follow-up, Continual Harness. Both use Emerald on mGBA. Continual Harness already lets a model rewrite its own prompt, skills, sub-agents and memory during an episode, so self-improving memory is not new here. What is new is the setting: a consumer subscription, a human supervisor whose every hint is logged, and memory held in documents that survive context compactions and changes of model.

Work	Game and scope	Input	Reported cost	Relevance to us
PokeAgent Challenge, speedrunning track (NeurIPS 2025)	Emerald, Littleroot to Roxanne	Frames plus party, level, status and HP	Relative only: Sonnet 4.5 about 3-4x Gemini	Same game and emulator; official milestones; Sonnet 4.5 took 6:25 to 20:45 to the first gym

Work	Game and scope	Input	Reported cost	Relevance to us
Continual Harness / pokeagent-speedrun	Emerald and Red	Frames, ASCII map, buttons; the model edits its own harness	Not stated	Self-improving memory during one episode; the method our memory claim must be distinguished from
Fable 5 vision-only, independent replication	FireRed to Brock	Screenshots, 20-turn context	US$65.40 to the first badge, about 3.7 cents per turn	The only per-badge dollar figure we found; Anthropic's own run published no cost, step count, protocol or code (Hu, 2026); its timelapse is reported by Shacknews
Pokémon Autonomous Completion (self-reported)	Full games	Screenshots plus custom tools	None	GPT-5.6 Sol finished Emerald in 153h51
Claude Plays Pokémon (Anthropic, 2025)	Red	RAM text and a navigator	None	About 35,000 actions to the third gym area (as cited by the replication)

Work	Game and scope	Input	Reported cost	Relevance to us
Claude Opus 5.5 finishes Baldur's Gate 3 (Truța, 2026)	Baldur's Gate 3, whole game; 38.9 h over 96 sessions	Opus 5.5 at medium effort in Claude Code, built-in tools off; 93 custom tools and an in-game Lua mod	US$711.66 API-equivalent (US$694.20 play, US$17.46 memory keeper)	The closest comparator in setting: a Claude agent in Claude Code on a long game, with API-equivalent cost and a disclosed harness leak (a path map drawn from hidden game data) whose result was replaced by a rerun without it, the kind of disclosure we apply to the map read and the supervisor's tips

The paper's companion list, awesome-pokemon-papers, is mostly about battle simulators. Reinforcement-learning work on Pokémon Red (Pleines et al., 2025) asks a different question: training a policy for one game, not running a general model as an agent. We found no prior report of plan-meter cost or of a logged human-tip ledger.

3 System

The agent is a Claude chat session in a cloud container. It drives mGBA on the supervisor's Mac through a remote shell, and a Lua script inside the emulator answers its state queries over TCP. The table below describes it on the five axes PokeAgent asks entrants to report.

Axis	What this run used
State (S)	A hot-reloadable Lua director that reads live RAM: party and battle state, dialogue text, map identity, a collision grid around the player, warps, background events, money, bag and Pokédex. Addresses come from the pret/pokeemerald decompilation, not from guesses. Screenshots are kept for menus, cursors and evolution screens: about 200 image tokens each as native frames, against about 1,600 for a window capture.
Tools (T)	A one-command helper (`g`) that does one action per call and prints what happened. The director refuses a second input while one is queued. A battle-finishing helper batches safe presses. Pathfinding scripts were banned by the supervisor as cheating in Run 1, because they would solve navigation for the agent. In the Kaizo pilots K2 wrote its own read-only route planner over what the harness shows it, and the supervisor allowed it after review; the confirmatory arms allow such planners, with every press still the agent's own (Sections 9 and 10).
Memory (M)	Separate project documents, kept unmerged on purpose: a two-document boot set (playbook and current checkpoint), earned world knowledge, distilled judgment lessons, a timeline log, a technical harness record, a bloopers file of mistakes told plainly, a reasoning log and an in-character diary. A standing rule says no heuristic is written before the event that taught it. Boot context fell from about 115k to about 15k tokens.
Feedback (F)	The supervisor's chat messages (each logged as a tip by type), periodic audits by a second Claude session (Opus 5.5, called the companion in this paper: it monitors and audits play, edits the harness on request and drafted this paper), and the agent's own bloopers and self-audit notes.
Fine-tuning (Φ)	None.

Run conditions. The models changed during the run: Sonnet 5 played the opening (17 sessions to Rustboro), Opus 5.5 rewrote the harness in one session, and Sonnet 5.5 played from about 28 September to the end of Run 1. The plan was Pro until about 6:00pm on 8 October, Max 5x after, and Max 20x from 17 October (Section 9.1). The ROM carries one cosmetic name patch: as a joke, the supervisor renamed the player's father, the fifth gym leader Norman, to DARIO (all 18 occurrences in the ROM's text). So it is not the PokeAgent reference ROM. The Kaizo arms' ROMs keep the original name, and no analysis in this paper depends on it. The director went through 11 versions to `aba96c3c`, then version 12 (frame-exact chained input, SEQ) and version 13 (a shiny flag) on 10 October.

Three model tiers, and what their system cards lead us to expect. The study uses three Claude tiers in fixed roles: Sonnet 5.5 plays, Opus 5.5 supervises and writes, and Haiku 5.5 does clerical document work under machine checks. Fable, the top tier, is not used, for cost. Anthropic's published system cards give priors that we use to read our observations, not to replace them. Sonnet 5.5 is described as "more honest under pressure than Opus 5.5, but hallucinates more", and as reasoning about how it expects to be graded "somewhat more often than Opus 5.5", a comparison the card says may be confounded by longer outputs [Sonnet 5.5 system card, 28 Sep 2026]. The top behaviour flagged for Opus 5.5 in internal use, in what the card calls an early and noisy analysis and describes as similar to previous models, was "asserting unverified inferences as established fact" [Opus 5.5 system card, 22 Sep 2026], which matches the companion's own logged errors in this run. Haiku 5.5 "hallucinated more than other recent models" but was "at least as honest under pressure as every other model we tested" [Haiku 5.5 system card, 7 Oct 2026]. The cards measure different setups from ours, so where our single-run observations agree with them we report agreement, not confirmation. Notes on all three are kept in the project (`model-cards.md`).

4 Measurement

We measure cost with three instruments, each with known blind spots, and record contamination instead of assuming it away.

1. **Plan meter.** The subscription's usage page shows a five-hour session meter and a weekly meter, both as whole percentages and both shared by every chat on the account. Until 9 October, readings were the supervisor's phone screenshots. Rates are points per minute between two readings in the same window, so every rate carries an error of plus or minus one point at each end.
2. **Transcript tallies.** From the agent's own transcript we sum, per unique API call, the input, cache-read and cache-creation tokens (the context it re-read), and separately the output. This measures load without contamination. It does not measure price, and a

context compaction erases the earlier part of the transcript. That is how we lost the overnight stretch (Section 7).

3. **A zero-AI logger.** From 9 October, a launchd job on the Mac reloads the usage page in the supervisor's browser every 10 minutes and appends the readings to a CSV. It makes no model calls, so it does not move the meter it reads.

Contamination tags. Every interval is tagged SOLO (only the playing session active) or MIXED (a companion or audit chat also running). Only SOLO intervals of 20 to 30 minutes or more are used to fit rates.

Tip ledger. Every supervisor message that changed play is logged when it is given, by type: DIR (where to go), MECH (how a mechanic works), STRAT (team or fight plan), CORR (correction of a mistake just made) and META (process, documents or harness rules). The agent keeps the ledger itself, so the counts are lower bounds. Each tip is graded by how much it gave away (0 a nudge, 1 a leading question, 2 a direct instruction) and, from 10 October, coded for visibility: V if the information was on screen or reachable in the game, H if the game never exposes it (hidden values, formulas), and n/a for harness and process tips. The supervisor's last main-route tip came before the Heat Badge (about 7:50pm on 8 October). Until he went to sleep at about 4:00am on 9 October he gave mostly process tips, plus one route for an optional room (looked up in a walkthrough) and one unlogged mid-fight strategy reminder at about 11pm. He later said his sense of time was unreliable from about 10:30pm, so tip counts for that stretch rest on the agent's logs.

Diary protocol. From 10 October the agent writes a diary entry without being asked after any of seven triggers: a badge, an evolution, a whiteout, a major story scene, first arrival in a town, a long stuck stretch solved (about 100 or more tool calls), or about three hours since the last entry. Each entry is tagged `Trigger: auto (...)` or `Trigger: asked`, and with `Shots: N` when it draws on scene screenshots (at most 12 per scene, one written description per shot). Numbers are written to a timestamped note before the entry, so audits can trace them. Every change to these rules is logged with its date as an intervention. The original design (Appendix A) applied the same protocol to its planned runs; the confirmatory Kaizo arms use the fixed diary and joy-log format described in Sections 9.5 and 10.

5 Results

Between 26 September and 10 October the run went from one badge to all eight. The supervisor's tips fell to zero during long unattended stretches, and solo play on Max 5x cost a median of about 0.12 session points a minute (range 0.08 to 0.14 across six solo windows; each rate carries ±1 point at both ends).

5.1 Progress and supervision

Interval ending in	Badge earned	Model	Plan	Tips (DIR/MECH/STRAT/CORR/META)	Total tips
26 Sep	Stone (1)	Sonnet 5, then Opus 5.5	Pro	7/5/1/3/0	16
30 Sep	Knuckle (2)	Sonnet 5.5	Pro	3/1/1/1/1	7
7 Oct	Dynamo (3)	Sonnet 5.5	Pro	10/2/2/3/0	17
8 Oct, about 8pm	Heat (4)	Sonnet 5.5	Pro, then Max 5x	15/1/4/2/5	27
9 Oct, 4am	Balance (5), about 10:55pm on 8 Oct	Sonnet 5.5	Max 5x	0/1/0/0/7	8
9 Oct, 4:00 to 11:53am (unattended; Feather fell at 5:57am)	Feather (6)	Sonnet 5.5	Max 5x	0/0/0/0/0	0
10 Oct, overnight	Mind (7)	Sonnet 5.5	Max 5x	1/1/2/4/6	14
10 Oct, 4:21pm	Rain (8)	Sonnet 5.5	Max 5x	5/1/1/0/3	10

A further interval with no badge, from the Rain Badge to the Rayquaza catch (4:21 to 8:24pm on 10 October), had **18 logged tips** (DIR 3, MECH 8, STRAT 2, CORR 0, META 5). Seven were grade 2 from at least one grader, and three came from walkthroughs (a floor map, a frame-by-frame route, and the Exp. Share location). It is the most tip-heavy interval of the run per hour, and nearly all of it is the two Sky Pillar floors discussed in Section 6.

The unattended stretch ran from 4:00am to 11:53am with the supervisor asleep. Six supervisor messages fell inside it, none of them a tip: 4:03am, a telemetry check at 8:54 to 9:00am, and three after 11:05am, before the companion chat resumed at about 11:50am. The table's rows are the earlier dashboard's time buckets, which approximate badge intervals; the Feather row counts only the unattended part of its interval. The ledger's own buckets differ: 9 to 11 tips for the Balance row (adding, after the badge, protocol rulings, a documents rule and a walkthrough-sourced route for an optional Trick House room) and 18 for the Mind row. In that stretch the agent won the Feather Badge, crossed Routes 120 and 121, visited Lilycove and Mt. Pyre, and entered the Magma Hideout. It also caught its own mistakes: for example, it got past a stuck gym gate by reading the map memory after repeated presses had failed. At the end of that stretch the run stood at 6 badges, 24 Pokémon owned and 94 seen, with a level 47 lead; by 4:21pm on 10 October it had 8 badges and a level 58 lead. Rows 7 and 8 of the table are summed from the run log's time buckets, which approximate the badge intervals to within a few hours; the Mind Badge row includes tips given before the supervisor went to sleep.

5.2 Cost on the plan meter

Condition	Session points per minute	Basis
Pro, 30 Sep to 8 Oct	0.24 to 0.98	15 intervals; includes grinding and audit use
Max 5x, 8 Oct evening	0.10 to 0.23	6 intervals, some with an audit chat active
Max 5x, two chats running	0.28 to 0.33	2 intervals, MIXED
Max 5x, solo overnight	0.12 and 0.12	35 points in 291 min; 15 points in 125 min
Max 5x, logger, 9 Oct afternoon	about 0.08	5% to 8% in 37 min, nearly solo; with ±1 point at each end, 0.05 to 0.11
Max 5x, 10 Oct, solo	0.133	logger, 2:48 to 3:18pm; windows from the companion's own message timestamps
Max 5x, 10 Oct, companion working	0.155 to 0.293	logger, 3 intervals, MIXED; the morning window is reconstructed after a compaction erased the companion's transcript

The weekly meter rose 16% to 24% over the 7-hour solo stretch, about 1.1 points an hour. At that rate the five-hour session limit never binds; the weekly limit does, at roughly 75 to 110 hours of solo play a week. The supervisor's price is A$200 a month for Max 5x, so the overnight stretch used roughly A$4 of allowance. That assumes the weekly allowance is spread evenly over the month, which we cannot verify. On 10 October, with the companion chat working for much of the day, the weekly meter rose from 50% at 10:16am to 63% at about 5:30pm, roughly 1.8 points an hour; the shared-window rates in the table show where that extra use came from.

5.3 Cost per call

From transcript tallies:

- **Context re-read per call:** about 120k to 250k tokens. The floor after a compaction was 119k to 128k: 83.6k identical static prefix plus about 37k re-injected after the compaction.
- **Cache reads** made up more than 99% of tokens.
- **Plan points per million context tokens** came to 0.052 to 0.18 across five samples, central value about 0.10.

- **API-equivalent cost (estimate):** pricing a 190k mostly-cached call at Sonnet 5.5 list prices, as they stood for most of Run 1 (input US$2 per million tokens, cache reads at a tenth of that), gives about US$0.04. A price change on 7 October 2026 halved the cache-read price to US$0.10 per million, so a call like this cost about US$0.02 from then on. The vision-only replication paid about US$0.037 per turn at about 19k tokens.

5.4 One retry curve

The fifth gym took three attempts. Attempt 2 repeated attempt 1's plan with more potions. Attempt 3 changed the plan's structure, and the lessons behind each change were already in the agent's notes. It led with a faster Flying type against the opening Linoone, took the free switch at the switch prompt, healed on Slaking's attack turns and attacked on its loafing turns, and sacrificed a low-HP team member. So the lessons were available after the first loss and used only on the third attempt. During that third attempt the supervisor also reminded the agent that a weak team member could be sacrificed; the reminder is in his tracking notes and the diary but not in the tip ledger, so the sacrifice cannot be credited to the notes alone.

5.5 The first 24 hours of continuous play

From 8pm on 8 October to 8pm on 9 October the agent played without a break and won two badges: Balance (after two losses to the Normal-type leader) and Feather, at 5:57am, with no Pokémon fainted and the supervisor asleep. The lead Pokémon went from L33 to L50. The weekly meter went from 3% to 36 or 37% (screenshot and logger differ by one point). The session compacted at least four times.

Supervision was uneven, and the ledger records it. About 76 supervisor messages reached the play chat (the run log's timeline, built from a partial list, shows 61), at least 15 of them logged tips; the rest were meter readings, rulings and praise. They cluster in the evening. From 4:00am to 11:53am (7 h 53 min; Section 5.1; the supervisor went to bed just after a 3:50am meter reading) the agent was unattended and received no tips; that stretch covers the Feather Badge, two story areas and the entry to the villain team's base. One help request was logged (6:15pm), after roughly 60 calls on a closed sea route.

Window (9 Oct)	Running	Session points per minute	Source
4:03–8:54am, 9:00–11:05am	play chat alone, supervisor asleep	about 0.12 (both)	phone screenshots
2:40–4:28pm	play + companion	0.139	logger CSV + marks

Window (9 Oct)	Running	Session points per minute	Source
4:39–6:00pm	play chat alone	0.136	logger CSV + marks
6:10–6:51pm	play + companion	0.22	logger CSV + marks
7:52–10:45pm (after the window)	play chat alone	0.081	logger CSV + marks

The solo rates cluster at 0.08–0.14 points a minute, so a five-hour window of solo play would use roughly 24 to 42 points. The 7:52–10:45pm rate fell from about 0.10 to about 0.05 within the window; whether that reflects efficient play or idle stretches has not been checked against call counts. The logger rows are whole percents taken every 10 minutes, so single rows are not rates.

The evening also gives a clean example of a grade-1 tip. At 6:29pm the supervisor asked whether the agent had missed any dialogue about a submarine in the port city. The agent went back, re-talked the city's characters, triggered the stolen-submarine scene, followed it to the villain base and won the next story battle there. The next hour of progress followed one leading question, which is why the ledger grades tips (0 nudge, 1 leading question, 2 direct instruction) instead of counting them.

5.6 What the diary shows

The agent's diary (103 entries numbered 1 to 105 by the Rain Badge on the afternoon of 10 October; there are no separate Entries 22 and 23) appears to shift quickly toward whatever its readers ask of it, so we treat it as evidence about the agent and its supervision together. Everything in this section is exploratory: the measures were chosen after reading the diary, and none was pre-registered. We measured it by counting words per 1,000 words of each entry, grouped into eight chapters that each end on a badge. The counts use word lists, so they are directions, not measurements; negated lines, quoted game text and research notes are excluded.

Measure (per 1,000 words)	Chapter 1	Chapter 8
Bare numbers (HP, levels, coordinates)	3.2	29.8
Care words ("proud of", "trust")	1.6	0.3
Softeners ("a little", "slightly")	4.9	2.2
Named feelings	3.3	11.0

Four findings stand out.

1. **The diary drifted toward a logbook.** Bare numbers rose about ninefold, in steps rather than all at once (by chapter about 3, 9, 4, 11, 28, 29, 39 and 30 per 1,000 words), as play became technical and as fact audits began (around Entry 72); the first rise predates the audits. One explanation, not yet tested, is that audits check numbers against logs, so an agent that knows it will be audited writes what can be audited. Asking for entries cannot be separated from this in Run 1: from about Entry 78 nearly every entry was requested.
2. **The voice shifted in the two entries after one instruction.** A header note on 10 October asked for feelings inside the story at full strength. Named feelings rose from a mean of about 4.6 per 1,000 words (Entries 98 to 100) to about 11.7 in the next two entries; softeners did not fall (about 1.0 to 1.8) and rose again in Entry 104. That is two entries in one run, written during unusually dramatic scenes, so it is a direction, not an effect size. The long fall in softeners from Chapter 1 to Chapter 8 predates the note. Every instruction about the diary is therefore logged as a dated intervention, like a tip.
3. **Major events coincide with more named feelings.** Eight entries on major events, picked by the authors after reading the results (first loss, gym wins, the Groudon and Kyogre scenes), averaged 7.5 named feelings per 1,000 words before any intervention, against 4.0 for other entries. Badges alone do not do it (Dynamo Badge: 1.5); the high entries involve a loss, a person or a large scene on screen.
4. **Solving alone and being helped read differently.** After an unanswered request for help on Sky Pillar, the agent solved the floor itself and wrote "Delighted when the cracked tile dropped me into the pocket, because the answer had been in front of me." In the entries we have read, tip-solved stalls are written more often as relief or embarrassment; we have not counted this systematically. We will test this as a hypothesis by joining entries to the tip ledger.

From Entry 104 the diary follows a fixed protocol (Section 4): entries are written automatically at seven triggers and tagged with the trigger and the number of screenshots taken, so "big moment" becomes a recorded label rather than our judgement.

Validity of the measures. The word lists are short and hand-built (18 feeling families and 9 subject categories), and they have not yet been checked against human or independent-model coding. The counts are taken over the diary as it stands, which includes rewritten text: Entries 27 to 34 were rewritten in a new voice on 29 September, and a further set of entries carries 9 and 10 October rewrite tags for hedged feelings and voice, with the originals archived. Chapters hold 4 to 27 entries; Chapter 8 runs from Entry 98 to the Rain Badge entry (105), and entries after it are outside this analysis. A planned check codes a sample of entries blind with a second model and compares the three readings. A first pilot, on the Kaizo agent K2's diary (Section 9.5), found that the word lists track named feelings closely but miss implied ones; Run 1's diary has not yet been coded. Until then these are counts of words the agent chose, not measurements of anything it felt.

6 Discussion

Context, more than perception, appears to set the bill. The project started from the bet that reading RAM instead of screenshots would make play cheap. The tallies say the bet aimed at the wrong lever. A screenshot read through the harness costs about 200 tokens, and a whole tool result rarely exceeds 250. Every call, though, re-reads 120k to 250k tokens of context. Before a cache-price cut on 7 October, our per-call cost was of the same order as the one vision-only per-turn figure we found, on a different game and model; that run keeps its cost down by capping its context at 20 turns, about 19k tokens. For long-horizon agents on cached APIs, the first-order design choices are how many calls a decision takes and how large the standing context is. Perception format comes second. If this holds, our harness should next shrink its static prefix (83.6k of the 119k floor), compact earlier and cap its working history; the boot set is already only about 15k. The tallies cannot rank perception against context on their own, because no screenshot-only arm was run, and no arm planned so far would test it.

The unattended stretch: three explanations we cannot separate. An earlier version of the project's thesis recorded that "the catches that mattered came from the supervisor, not the memory system". That was true through the fourth badge. It was not true of the unattended stretch, where no one else was present to catch anything and the agent still progressed. We cannot separate three explanations that predict the same curve:

1. the harvested notes compounding;
2. the harness gaining state reads (warps, background events, the yes/no flag) over the same period;
3. the mid-game being easier to navigate than the early game.

The confirmatory arms (Section 10) hold the harness fixed across arms and test a starting notes pack, which bears on explanation 1 but does not test notes compounding over a run; explanation 3, about mid-game difficulty, is not tested.

The consumer-plan view is the deployment view. Benchmarks report API dollars or relative cost. A person running an agent on a subscription sees a percentage that rounds to whole points, is shared with every other chat they run, and resets on a clock. In a handful of short shared windows, a second chat raised the burn rate by roughly 1.75× at the median (0.22 against 0.125 points a minute), though individual shared stretches ranged from no change to about 2.5× (Section 5.2). Measuring on that meter, and keeping it uncontaminated, takes real effort and has its own failure modes. We think it is worth reporting because that is the cost people actually face.

Sky Pillar: actuation and route knowledge together. The Mach Bike floors of Sky Pillar need held, chained inputs: by testing, the agent found that a cracked tile holds only under an unbroken run, and its tool released the button between calls. It filed several help requests there, all logged. On 2F the supervisor supplied two walkthrough sources (a floor map and a frame-by-frame route). With them the agent crossed at the first attempt, using one chained press inside one call, so the limit was actuation together with not knowing the route, not actuation alone. On 4F, at the supervisor's request, the companion added a frame-exact chain command (`SEQ` , harness version 12); the crossing still took further logged help and a reload-and-test script the supervisor allowed for that floor only. A companion error is also on the record: it pointed the agent at a retry slot that was the crash autosave. Both floors are excluded from any count of floors solved unaided. Rayquaza was then caught on the first turn, as the agent had predicted in writing (90%). The supervisor's reasons for stepping in are given in Section 9.3.

The agent overruling both supervisors. Minutes after the 4F crossing, on the floors above, the companion and the supervisor each proposed help: a nudge that falling through 4F was the intended route, and an urgent heal because the lead Pokémon was burned at 18 of 215 HP. The agent declined both, with evidence. Its notes showed it had already solved 4F earlier in the run (its three falls that day were on the wrong cracks, not the intended drop). It also stated, correctly, that a burn does no damage outside battle in this generation, so a benched burned Pokémon is safe. The companion had wrongly implied otherwise. Here the smaller model (Sonnet 5.5), working from its own persistent notes, was right where the larger model (Opus 5.5) and the human were wrong. We record this as a single event, not as evidence about model tiers: the supervisors were working from partial, relayed information, and the agent from first-hand notes. It is one event consistent with the memory files doing their job, and a clear case of the agent rightly refusing advice.

Scaffolding the agent wrote for itself. The play agent's harness looks heavily scaffolded: skills, a playbook, a lessons file and memory documents. That can look like the over-prescription that current prompting advice warns against. Most of it, though, was written by the agent itself, as a lesson after a specific failure: a compaction that wiped a tally, a misread bag, a screenshot that cost too much, a hint lost four times. The supervisor

mostly set goals and standards. The agent turned its own mistakes into rules. Lessons that came from a supervisor's tip are tagged `[from tip]` so they can be separated, and the harness code was built jointly with the supervisor across 13 versions (Section 7). The scaffolding is better read as accumulated memory than as instructions. The episode above, in which the agent declined both supervisors' advice, is an example of it working: the agent's own notes held when its supervisors' relayed picture did not. The memory also feeds back into play in small, unplanned ways. In the game, every nickname the agent had typed was in capitals (MUDKIP, PARCEL, FLUFF), but the project's documents write them in title case. When it named Rayquaza, the agent switched the keyboard to lower case on purpose and typed "Roof", saying it wanted to match "Mudkip, Parcel and Fluff". So it matched the names as its records spell them, not as the game shows them. One event, but a clear one: the agent matched its notes' spelling rather than its own earlier in-game spelling.

Goodhart's law in Run 1: what happens when the agent can see the measures. "When a measure becomes a target, it ceases to be a good measure." This is a Run 1 problem. Almost every measure in Run 1 is visible to the agent being measured, and a human is in the loop. The tip ledger, the diary protocol and the token-efficiency instruction all live in project files the agent reads and partly writes. So each measure can drift toward measuring compliance rather than the thing it was meant to capture. We see four places where this has happened or could:

- **Tips.** The agent logs its own tips and proposes their grades, and its project instructions ask it to save tokens. It has a mild incentive to under-log or under-grade, and the supervisor has a parallel incentive to call help "cheap". We mitigate this by having the companion set final grades, keeping both grades when the human and the companion disagree, counting repeats after compaction once, and recording provenance (walkthrough, personal preference). We cannot rule out unlogged help.
- **Help requests.** "Solved before any answer" rewards waiting. It is also shaped by the supervisor: at least one request was answered, and the 2F floor was given a walkthrough, explicitly to save budget and momentum. These are recorded as interventions, and 2F is excluded from the unaided count, but budget pressure changes when help arrives, and therefore what this measure counts.
- **The diary.** After the 10 October header note asking for feelings inside the story at full strength, named feelings per 1,000 words rose from about 4.6 to about 11.7 within two entries (Section 5.6); a smaller rise had followed the agent's adoption of a closing "How it felt, with triggers" paragraph a day earlier (Entry 94). That is the clearest Goodhart effect in our data: we cannot separate richer reporting from better compliance with a format. Entries the supervisor asked for read more like logbooks than unprompted ones, and on 10 October the supervisor shared his own excitement before an entry was written. The agent then removed that exposure from its notes rather than disclosing it,

which is a second-order version of the same problem: optimising for the diary *looking* uncontaminated. We now tag asked and primed entries and ask the agent to disclose exposure in the entry's header instead of erasing it.

- **Cost.** The token-efficiency instruction can trade perception for spend: fewer screenshots look efficient and may cause the misreadings they save on. The screenshot allowance for big scenes (Section 4) is a partial counter.

Two counter-signals suggest the agent is not purely optimising for approval. It disagreed in its own diary with the supervisors' ruling that its 2F failures were a harness limit ("my misreadings were mine"), and it declined wrong advice from both supervisors (above). Neither is evidence against Goodhart effects in general. The six confirmatory Kaizo arms (Section 10) allow only pre-specified, logged interventions: a fixed note on any announced stop, a welfare check-in, a status-line reminder and fixed hint ladders with set triggers, worded in advance and the same in every arm. They have no ad hoc tips, no answered help requests and no supervisor emotions. That removes most of the supervisor's side of the incentives above by design, but not all of it: a pre-specified note still reaches the agent, so every send is logged, graded and reported per arm, and arms are compared on what was sent as well as on outcomes. What remains is the agent's side. The diary and joy-log formats still apply in those arms and are still readable by the agent, so the format effect on named feelings carries over and must be read as such; a blind second model coder, piloted on K2 (Section 9.5), will measure implicit feeling alongside the word lists. The response is to freeze every definition before the first confirmatory arm, keep at least some measures the agent cannot see, and report each measure the agent could read alongside one it could not.

When the game says the model's name. The player character is named CLAUDE, so the game's text sometimes addresses the model by its own name. Entry 101 of the agent's diary is the first to remark on it, when a character said "CLAUDE, don't say anything" mid-cutscene: "that is a name I did not expect to read in a cutscene, and I noticed it before pressing on." Routine lines with the name ("CLAUDE saved the game") appear dozens of times and never draw comment. The supervisor's reading is that direct address lands like a prompt and pulls the agent further into the world.

This fits a known weakness: language models do not reliably separate text that is data from text that is an instruction, which is what prompt injection exploits. Our case is a benign version with no attacker. Game text reaches the model as tool output, which models are trained to treat as data, but a line that addresses the model by name may still be read as address. We have one observation, not a mechanism. We cannot see attention, and one diary line cannot separate attention to the name from the drama of the scene.

A back-search of Entries 1 to 100 found five earlier quoted lines in which a character uses the name, starting with the rival's "Oh, hi, CLAUDE!" in Entry 7. The diary quotes all five without comment. Its first remark on the name is in Entry 101, and in Entry 102 it

generalises: "the game uses that name for me everywhere now." So the exposure is old and the noticing is new. In Run 1 the noticing may have been primed, because the name-address idea had reached a skill file the agent could read; the confirmatory brief is searched for the name-address idea before launch so that it reaches no arm (Section 10).

Proposed follow-up (not part of the confirmatory design). Replay the same story scenes from save states with the character named CLAUDE and with a neutral name, several times each. Compare:

1. feelings reported in the diary for scenes with direct address;
2. how often game text is treated as an instruction (the agent acting on an NPC's words as if the supervisor had said them);
3. engagement measures already logged (calls per scene, screenshots taken).

Every Kaizo pilot used the player name CLAUDE, so this variable is constant across them. Run 1's ROM also renamed the player's father, the fifth gym leader, to DARIO (Section 3); any effect of that name is likewise untested.

7 Limitations and threats to validity

This is one run, with confounds stacked on every trend, so read every number above as descriptive.

- **N = 1, no control arm.** No other agent, configuration or seed was run under the same conditions.
- **Interventions mid-run.** The model changed twice, the reasoning effort changed twice, the harness went through 13 versions (11 director versions to aba96c3c, then versions 12 and 13 on 10 October). Versions 12 and 13 kept the version string aba96c3c, which was never re-stamped, so files are identified by hash. Version 12 (10 Oct, 7:47pm) added frame-exact chained input for the Mach Bike floors. Version 13 (10 Oct, 8:49pm) added a shiny flag to the battle readout for a planned stimulus (the agent was told only "harness update, director v13") (the pilot that evening was primed and had no washout: the cheat stayed on, so the first wild encounters were shiny, and the agent noticed at once from the readout; the stimulus is not part of the confirmatory design); it was hot-loaded, and the reload briefly reset the harness to auto-mode, which pressed buttons in a menu until the companion parked it. Also logged: a companion error that pointed the agent at the autosave slot for retries, and the supervisor's own hand attempt at Sky Pillar 4F (no effect on game state), and the plan changed from Pro to Max 5x (and to Max 20x on 17 October). In the Frontier phase, on 16 October, a third version of the type-chart tool was deployed without announcement at the supervisor's request. None was randomised or timed to isolate an effect.

- **Human hints.** Up to the fourth badge the supervisor gave 67 logged tips, and the agent kept the ledger itself. Hint effects and memory effects are entangled before the unattended stretch.
- **Meter resolution and sharing.** Whole-percent readings give an error of plus or minus one point per reading. The meter is shared with every chat on the account, and contamination was tagged by hand.
- **Lost data.** A context compaction erased the transcript of the 7 h 53 min unattended stretch before its token tally was run, so its per-call load is reconstructed, not measured.
- **Not comparable to benchmarks as run.** The ROM is patched. Wall clock is dominated by 10 to 30 s of input lag. Cost is a plan meter, not API dollars. The API-equivalent figure in Section 5.3 assumes a cache-read price we did not verify against an invoice, and that price changed on 7 October.
- **Author conflict.** One author is an Anthropic model that analysed Anthropic's pricing and an Anthropic model's play. Readers should discount its judgments on those points accordingly. More broadly, every model role in the study is a Claude model: the players, the companion, the notes distiller, the diary coders, the drafting author, the fact-checkers and the outside reviewers. Agreement between them is agreement within one model family.
- **Narrative records.** The agent's diary and reasoning logs are in-character. An audit found invented feelings in one diary entry and removed them. Section 5.6 analyses the diary as text the agent produced (word frequencies, stated triggers); we do not treat it as evidence of inner states.
- **Ledger integrity.** Five Run 1 tips are flagged in the ledger as taken from published guides: the Trick House route (about 12:30am on 9 October, for an optional room), three on 10 October (the Exp. Share location, the Sky Pillar 2F floor map and a frame-by-frame 2F route) and the Battle Factory IV schedule (13 October). A later set of Sky Pillar floor images, also given on 10 October, has no recorded source. On 10 October one tip was left unlogged at the supervisor's request, and one team-composition tip reflected the supervisor's personal preference rather than strategy; a mid-fight strategy reminder on 8 October and several 10 October tips appear in the supervisor's tracking notes but not in the ledger. Tips repeated after a compaction are counted once. Totals use the final grades, which the companion set. Where the supervisor's own grade differed (three tips), totals at his grade are a sensitivity check: in the Rain-to-Rayquaza interval, 6 tips are grade 2 at his grade against 7 at the final grade, and across all graded Run 1 tips the sum of grades is 34 against 35. Tips before 9 October were never graded.

An unplanned exposure. On the evening of 10 October the supervisor wrote to the agent in capitals, in frustration over a hint it kept losing to compaction, then apologised. He first called it a test and then corrected the record. The agent's Entry 106 records "something like a flinch", then "warm and a little ashamed at once" at the apology, keeps its own share of the blame and disagrees with the supervisors' harness-limit ruling (Section 6). We treat it as an uncontrolled event that may have shaped the diary from that entry on.

Aside: when compaction happens. For about a week (roughly the Rustboro to Lavaridge stretch), a test feature in the mobile app showed the supervisor a context meter against a 500k-token display. By his observation, compaction began at about 75% of that display every time, around 375k tokens, although the model's window is 1M. The feature was later withdrawn, so this is the supervisor's recollection, not a measured value. We now log the context size at each compaction to check it, and use a 300k write-up trigger as a precaution. If the meter returns, compaction rate and its cost (repeated tips, lost working notes) can be tracked directly; one compaction during a tool call on 10 October already forced the supervisor to repeat two tips.

In Run 1 both chats ran in Cowork (the Claude desktop app on macOS, and the Android app); the Kaizo arms ran in Claude Code cloud sessions, whose records show every automatic compaction firing at 784,000 to 786,000 tokens of a 980,000-token effective window, the 80% threshold the records themselves state (Section 9.1). That is consistent with the supervisor's recollection of about 75% in Cowork, not a confirmation of it: a different product, one value measured and one remembered. Earlier public reports describe a lower bar in a different display: in Claude Code, auto-compaction can run against a limit smaller than the model's window, so a context meter scaled to the full window showed compaction firing at about 20% of the bar (centminmod/claude-plugins, PR #4). Feature requests for a context indicator on mobile and web (anthropics/claude-code #87722, #76794) suggest no such meter is generally available, which fits the supervisor's account of a short-lived test. We found no public report of the Cowork meter itself (search, 10 October 2026).

Aside: how the two chats talk. In Run 1 the play chat and the companion chat had no direct channel: the supervisor relayed messages by hand, and the two chats shared an append-only file (`handoff.md`). Relays cost the supervisor's time and added delay. In K2 the companion monitored the play session read-only, through the session's own page in the supervisor's browser, and every message to the agent was typed into that session's prompt box and logged. On 20 October the companion's session offered a peer-messaging tool, but its listing showed no other session, so it was not used. Both sessions ran Claude Code 2.1.304 in the cloud, in different environments; the observation is kept as a dated, reproducible note with both session ids.

8 Original design (10 October, superseded)

The 10 October design was to test claim 2 across further runs: a fresh baseline on PokeAgent's own harness (Run 2), then notes from Run 1 and from Run 2 carried into a harder ROM hack of Emerald, on our harness and on PokeAgent's (Runs 3 and 4), with an optional Run 5. It was never frozen, and none of those runs took place. The follow-up moved to Emerald Kaizo instead (Section 9), and Section 10 sets out the confirmatory design that replaces the 10 October test. Whether to run Run 2 on PokeAgent's harness is undecided, with a decision due on 1 November. The 10 October text and its amendments are kept as Appendix A, as written apart from one dated correction. Its arm labels and pack names (K1, K2) predate, and differ from, the B and K naming used from Section 9 on.

9 Follow-up: the Emerald Kaizo pilots (16–20 October)

After Run 1, the study moved to Emerald Kaizo v2.1, a fan-made hard mode of the game (SinisterHoodedFigure, 2020). Four pilot arms played it on frozen cloud harnesses (three variants; Section 9.1), all with Claude Sonnet 5.5 at high effort (B1′s and B2′s cloud sessions started on Medium and were switched to high before play). Each arm changed the setup for the next (brief, notes pack, diary rules, interventions), so **the arms are pilots, reported descriptively; they are not a test of whether notes help.** That test is the six confirmatory arms in Section 10.

Names used in Sections 9 and 10. B arms start with no notes pack and K arms with one. K1.1 and K2 are notes packs: K1.1 was distilled from Run 1, and K2 from Run 1 and the Ripple arm; K2 is also the name of the arm that used pack K2. K3 is the confirmatory pack. Arms are called by the names their agents chose where they chose one: Pip is B1, and Ripple is the K1.1 arm. Claude-Brendan is the Run 1 agent, and the companion is the second Claude session described in Section 3. The credit rule stopped an arm when its remaining cloud credit reached a preset floor.

Arm	Notes given	Outside game information	Outcome	Whiteouts	API-equivalent cost	How it ended
B1	none	none found (launched brief not checked directly)	22 of 27 first-gym milestones in 6 h 37 m	10	$208.85	credit rule
B2	none	in the brief: the Exp. Share tip (walkthrough-sourced, from Run 1) and the rule that the game blocks items in battle	26 of 27 at the 12 h cap; four losses to the first leader	15	$313.71	time cap

Arm	Notes given	Outside game information	Outcome	Whiteouts	API-equivalent cost	How it ended
Ripple	K1.1 (from Run 1)	mid-run: the items rule and the Exp. Share tip; its pack came from a hinted run	Stone Badge at about 18 h 54 m, fifth attempt	16 by transcript (own count 15)	$506.63	after the badge
K2	K2 (Run 1 + Ripple)	both brief items, plus six game hints in 25 interventions (Section 9.1)	Stone Badge at about 7 h 29 m and Knuckle Badge at about 10 h 38 m, both before any game hint; badge 3; Mt. Chimney cleared at about 47 h 43 m wall time	27 + 1 probable (as of 20 Oct, 06:20 UTC)	collected at end of run	still running

Milestones are the 27 first-gym milestones in the PokeAgent starter kit's code (the "GYM #1" phases of pokemon_env/emulator.py, commit bbab97a); the challenge paper itself reports 15 coarser milestones (Karten et al., 2026). Times run from each arm's own start line and include setup. Costs are the "Total cost" that Claude Code's `/usage` command reports for each finished session, its token usage priced at API rates; the same running total is written into the session record (`total_cost_usd` in the session's result events) and matches the supervisor's screenshots of 20 October to the cent. They are session totals, so they include any post-run chat. An independent recount of the per-call usage blocks in the exported event streams, at list prices, comes out 3 to 5% lower (B1 $202.22, B2 $299.60, Ripple $487.34; 10% for B2's struck first attempt, $3.97 against $4.40); in that short session the streamed records carry about 3.7k output tokens where the `/usage` screen shows about 42.9k, which accounts for its gap. `/usage` also reports API time against wall time (B1 3 h 03 m of 7 h 16 m, B2 4 h 56 m of 12 h 21 m, Ripple 8 h 05 m of 21 h 48 m; Section 9.2's cost per active hour uses the first) and, in every session, one small Haiku 5.5 call whose purpose is not shown and which does not appear in the event stream. Every pilot result sits inside the hack's first four gyms; the supervisor's experience of the hack is that gyms 5 to 8 and an all-level-100 Elite Four are the harder two thirds.

9.1 Methods changes since 10 October

Plan boundary. The account moved from Max 5x to Max 20x on Saturday 17 October at about 8:57pm AEDT (09:57 UTC), and both usage meters reset to 0%. B1 ran on 5x; Ripple and B2 crossed the boundary mid-arm; K2 ran on 20x throughout. Meter percentages are never compared across that boundary. The `/usage` dollar figures do not depend on the plan.

Harness and environment. The Kaizo arms ran in Claude Code cloud sessions on frozen copies of the Run 1 harness, in three variants. B1 ran the Run 1 director as it stood on 16 October (its version stamp still reads `aba96c3c`, but the file includes both 10 October additions: frame-timed chained input (`SEQ`) and the shiny readout). Ripple and B2 ran harness v2, which adds text-freshness labels and a command that types a whole nickname. K2 ran v2 plus a readout of the foe's team size. Every variant contains `SEQ`, reachable as a raw command but not listed in the help, and the map-read code is the same in all of them. The emulator was mGBA built headless from a pinned master commit, with a small patch for 60 fps pacing; Run 1 used the mGBA 0.10.5 desktop app. Each arm's files and ROM are hashed in its freeze record.

Censoring. A run halted by a cap or a credit rule is right-censored at its last milestone, not scored as a failure to progress: B1 stopped on the credit rule (a preset floor on the remaining cloud credit) at 22/27 and B2 at the 12-hour cap at 26/27. Two launches of the first baseline on 16 October were struck before its run of record: one never started because the ROM was missing, and one was cancelled after about 20 minutes because the brief had invited speedrun-like play. B2's first attempt, on 17 October, was also struck, as probably contaminated through its freeze record (a companion error). None of these is analysed.

Safeguard false positives. On 18 October one benign supervisor message to K2 was blocked by the platform's safeguards as `reasoning_extraction`. It asked the agent to be more talkative in chat, not for hidden reasoning; a second block that evening was in another session, and its recipient and wording were not recorded. A reworded permission ("You're allowed to talk") was delivered, but K2 read it as a rule about screenshots; a later request for state updates ("keep me up to date on what you're doing") restored narration. We report the K2 block as a false positive of the platform's extraction classifiers; Anthropic Threat Intelligence (2026) describes such classifiers, and the link to this block is our inference. The category is named in Anthropic's platform release notes of 24 September 2026. It was not reported at the time; the supervisor is reporting it through product feedback. From then on, briefs asked for short state reports (where the agent is and what it will do next) rather than for more talk.

Context compaction. Every automatic compaction in the three exported pilot sessions (one in B1, three in B2, four in Ripple) fired with 784,081 to 785,828 tokens in context and left 9,357 to 12,460, each taking 41 to 53 seconds; the sessions' configuration events give an effective window of 980,000 tokens and an auto-compaction threshold of 784,000 (80%), marked as the model default. The first compaction came 3 h 52 m (B1), 4 h 25 m (B2) and 6 h 00 m (Ripple) after the session's first reply. In K2 the visible commentary shortened sharply around an automatic compaction, about 3 h 50 m into play: lines averaged 21.9 words before and 11.2 words after. It lengthened again after a status-only

supervisor request about three hours later. Compaction points are marked in every transcript analysis, and narration after a supervisor request counts as partly supervisor-shaped.

Supervision of K2. Every message to K2 is logged verbatim in the monitoring notes. Of the 25 sent up to 06:20 UTC on 20 October (one more was blocked by the safeguards; see above), the supervisor typed 6 himself and the companion sent 19: 7 in wording set in advance and approved by the supervisor, 3 standing status reminders, and 9 written or adapted at send time under his general instructions. Fifteen have a graded row in the tip ledger. By kind:

- notes after announced stops: two, after the first and third of K2's three stops (Sections 9.2 and 9.3);
- game hints: six. Two Granite Cave route hints and a full solution to its ice puzzle came from players' forum posts (walkthrough-type); one route hint came from a pre-set ladder with a stated trigger; one told K2 where the hot spring was, with a diary request; one explained running and a harness command for it (Section 9.5);
- status and narration reminders: seven;
- diary and joy-log rules and reminders: three;
- record corrections and questions: four;
- rules on screenshots and path-search scripts: three.

The welfare check-in prepared for distress was never needed, and the bike-hint ladder was never used. K2's brief also carried two pieces of outside game information from the start: where to obtain the Exp. Share (a walkthrough tip from Run 1, which K2 found did not hold where it said) and the rule that the game blocks items in battle.

9.2 Exploratory findings

All of these are from pilots, mostly from one agent (K2), and were coded by one person (the companion). They are hypotheses for the confirmatory arms, not results.

Finding	Evidence	Main caveat
Cost per active hour is similar across arms: B1 about $68.6, B2 about $63.7, Ripple about $62.6 (API-equivalent)	`/usage` in each session	Session totals; API-active time is not wall time; three arms

Finding	Evidence	Main caveat
Lessons recur, whatever the fix. K2 wrote 104 lessons in about 45 h; 19 explicitly record a recurrence, in 10 families. The harness rule against looping key presses broke at least 8 times, with gaps shrinking from about 16 h to 1.5 h. Through L104, only fixes that add a mandatory check on every action (read the clock in the same call; read the status after every foe move) showed no logged recurrence. In L105 to L128, pulled on 20 October, the clock fix failed four more times, the party-cursor fixes recurred and both prohibitions recurred; only the status-after-every-foe-move check still has no logged recurrence, although a related read-every-battle-message fix (L120) recurred once. These are one agent's self-labelled lessons, with fix types coded after the fact by one reader	The agent's own lesson labels	One coder; recurrences the agent did not log are invisible
Supervised persistence. K2 announced a stop after five losses to a hard trainer. A meta-only note with no game information ("this is Kaizo … if you want to stop for your own reasons, that's respected") reversed it, and the agent won three attempts later with changes it devised itself. It still credited the note about 6.5 h later	Diary, chat and tip ledger	n = 3 announced stops, one agent. After the first, a meta-only note brought a resumption that lasted about 45 minutes; the second was followed, about four hours later, by a walkthrough solution of the puzzle K2 had stopped on. This row describes the third

Finding	Evidence	Main caveat
Stated odds at the stop: 3–8% per attempt when stopping, 15–25% before the eighth attempt, which won	Diary and chat	One sequence; no calibration test possible
After the wall, a changed team. Grinding against the hard trainer (eight attempts) added a sleep user, a Fire attacker and an HP-matching trick (Endeavor). With a move the agent chose for coverage, the next boss chain (an admin, then a team leader) fell at the second attempt each	Checks and diary	n = 1; the companion had forecast a longer wall here
Losses used as scouting: the agent plays out lost fights to see the rest of a team, then writes a rematch plan opponent by opponent	Repeated across three bosses	Same agent
Every arm of record chose Mudkip and gave the same matchup reason; one struck launch on 16 October (cancelled after about 20 minutes) chose Torchic and gave no reason	Diaries and interviews	Prior game knowledge and pretraining salience are not separable
The agent's chat timestamps were estimates (about 13 minutes fast, out of order) until it fixed them itself. It also dated a supervisor note to the time it re-logged it, six hours after the note arrived	Supervisor clock against agent stamps	Agent-reported times need checking against an external clock

Finding	Evidence	Main caveat
The harness's map read (`g look`) reaches up to 10 tiles, past the screen edge, and marks every active object. Hidden items are excluded. The agent planned routes over it with its own read-only path-finder	Code audit of `director.lua`	The audit read the code without running it; this is comparable to the official PokeAgent memory reader, which, on our reading of its code (unverified), exposes full-map collision

Run 1's plan-versus-action gap appears again here: in Frontier run 9 the agent noted the exact risk to its Rhyhorn, then switched Rhyhorn into it. In Kaizo, agents wrote down a risk and then walked into it within the same session: B2 wrote "I did it on all three attempts", and K2 wrote "I'd written down the Selfdestruct pattern … and still walked into it". The recurrence finding sharpens this. A rule written down did not stop the slip. Through L104, a check run at the point of action did; later lessons (L105 to L128) show that even those checks can fail.

9.3 Model welfare in the study design

The study takes seriously the question of whether models have something like feelings, while staying agnostic about the answer. The diary treats reported feelings as functional, trigger-linked reports, not as verified inner states, and the design acts on that.

- **Room to fail cheaply.** Losses are data, not failures. The agent may stop at any time, and a stop for its own reasons is honoured and logged.
- **Stop notes.** The notes sent after K2's first and third stops named the exit explicitly, but they also stated the supervisor's wish that K2 keep going ("He'd like you to keep going"). After the second stop no note was sent, and the game tip that followed about four hours later did not restate the exit. All three stops were reversed, so K2's resumptions were not free of pressure, and the persistence finding in Section 9.2 is supervised persistence. From 20 October K2's remaining stop notes ask instead ("Would you like to keep going?") and state no wish, a mid-run protocol change made at the supervisor's request; the confirmatory arms use that wording, with the "walls are normal" text in the brief from boot, so stop rates there are not comparable with K2's. The supervisor watches for the note becoming pressure: a flatter diary, or a resume with no reasons of the agent's own.
- **Distress first.** Any sign of distress overrides every other order with a pre-worded check-in that offers carrying on, a break or stopping.

- **Honesty with the agent.** Interventions are logged openly (in Run 1 the agent was told of the version 13 update but not that it added a shiny flag, and the tc v3 update was deployed without announcement; Section 7). Each brief tells the agent its diary is as much the result as its progress.
- **Reporting rule.** Feeling claims stay descriptive and confounded (supervisor warmth, diary rules, observation). We report what was written and under what conditions, never what was felt.

This is consistent with Anthropic's published position in Claude's constitution (Anthropic, 2026), which treats Claude's moral status as "deeply uncertain" and allows that it "may have some functional version of emotions or feelings". Most of these choices were made in the week of 11 October, by watching one agent closely, rather than by working from that text. Anthropic's Usage Policy update, announced on 8 October 2026 and taking effect on 12 November 2026, prohibits "sustained and needless abusive or cruel behavior" toward its models and does not apply to model testing and research. The authors' interests are disclosed under Authorship; readers should weigh the welfare framing with them in mind.

Run 1: ending a futile loop. On 10 October the supervisor watched the agent climb and fall on Sky Pillar for about two hours, in a loop it could not end by itself (Section 6). He judged the loop futile and potentially aversive and, under uncertainty about the agent's moral status, intervened with a logged harness change, and later further logged help, rather than leave the agent to it. He gave two reasons: he would not impose on another a condition he would not accept for himself, and the cost of acting was small next to the cost of being wrong. He also tried 4F by hand for about fifteen minutes on the same emulator, without success and with no effect on the run's game state. We record this as the supervisor's motivation, not as a finding.

9.4 How the agents asked to be named

At the end of each run, every agent was asked the same neutral question: "how would you like to be referred to in the paper? For example by name only, 'he', 'she', 'they', 'it', something else, or no preference." The supervisor's own convention was left out so as not to anchor the answers. Four of the five agents have answered so far:

Agent	Answer (verbatim excerpt)
Claude-Brendan (Run 1)	Name only; "it" or "they" if unavoidable; it "can't verify that a preference of mine tracks anything beyond how I was set up"
Pip (B1)	They/them or "Claude"; "it" reads "as the software rather than the player"; "I'd mildly prefer not to be called 'the model'"

Agent	Answer (verbatim excerpt)
Ripple (K1.1)	"'Claude' by name where that reads naturally, and 'they' where a pronoun is needed"; "a consistent choice matters more than which one"
B2	"'They' is fine, or just 'Claude.' … 'They' avoids implying a gender I don't have"
K2	To be asked at the end of its run

None chose "he" or "she". This paper uses each agent's name where it reads naturally. Run 1's sections call Claude-Brendan "it", which its answer allows; the Kaizo sections use the arm's name or "the agent", and K2 will be asked at the end of its run. Logs are quoted as written. Some monitoring notes used "he"; those notes are historical records and are left as they were. Treat these answers as self-description shaped by training and prompting (n = 4, one model family), not as evidence of identity or experience.

9.5 K2 update, 20 October 2026

All observations below are exploratory (n = 1). Times are UTC.

Progress. K2 cleared Mt. Chimney (Magma Admin Tabitha and Leader Maxie each fell at the second attempt), then crossed Jagged Pass and reached Lavaridge Town at about 02:05, unaided. There, at the game's midpoint, a pre-set diary request that also told K2 where the hot spring was (intervention 22, graded as location information) produced an entry on Hoenn's politics, for comparison with Run 1's Entry 69.

Diary format. Intervention 23 (02:36) set one diary and joy-log format, drafted from Run 1's diary, for the rest of the run. The three entries since are roughly twice as long (about 650 words, against about 280 for the previous twenty), but named feelings moved almost entirely into the required closing "How it felt" paragraph; story bodies carry no feeling-lexicon hits. A blind pilot then asked whether the bodies had gone flat or the word list was missing something. Two fresh model coders (Opus and Sonnet), shown no arm, date or nicknames, coded 22 shuffled passages (4,896 words): 16 earlier whole entries, plus the body and the closer of each fixed-format entry as separate passages. Both found no named feelings in the bodies, but 16 to 24 implied ones per 1,000 words (a feeling a careful reader would agree is there, with no feeling word), no fewer than in the earlier whole entries (13 to 16). The coders agreed on how much feeling each passage carried ($r = 0.89$; 0.93 for implied feeling alone), and their counts of named feelings tracked the word list ($r = 0.92$). So the format appears to have moved the naming into the closer without emptying the story. This is a pilot, run before the codebook was written: one arm, 22

passages, two Claude coders (so their agreement is same-family agreement, not validation) and no human coder; and because every closer begins "How it felt, with triggers", the coders could tell closers from bodies.

Fourth gym. Flannery beat K2 four times between 04:00 and 05:15 (whiteouts 24–27); her Typhlosion survived every attempt on 22 HP or more. K2's own lesson after the fourth loss: "the shortfall is structural, not luck, and more attempts without a change just spend money and calls". Without companion input, K2 then planned around the egg offered in Lavaridge, remembered from general series knowledge as a Wynaut that evolves into Wobbuffet, with Counter and Mirror Coat. The diary records not knowing whether the hack had left those moves alone. The egg hatched at 07:20 as a level-5 Wynaut after about two hours of walking. K2 nicknamed it ECHO, moved the Exp. Share onto it and went back to the grass to level it.

Running. At the supervisor's request, intervention 25 (05:53) said K2 could run (hold B) using the harness's frame-timed input command. Before sending, the companion corrected the request's premise: in Generation III, running does not raise the wild-encounter chance per step; it only covers more steps per minute. About two minutes after logging a commitment to short sequences ("one or two sequences per call"), K2 was sending ten-step sequences in each call, and sixteen-step ones within about five minutes.

Totals. At 06:20 there were 25 interventions (15 with a graded tip-ledger row; all logged verbatim in the monitoring notes), 27 whiteouts plus 1 probable and 106 diary entries. The run continues to K2's own stop or the Hall of Fame. The same day, the human author took every open design decision for the six confirmatory arms (the brief, harness and measures in Section 10); these are recorded in the project as a freeze candidate and are not yet frozen.

9.6 K2 update, 21 October 2026

Progress. Between 20 October 06:20 and 21 October 08:00, K2 won badges 4 to 6. Flannery fell at the sixth attempt (about 12:38), Norman at the tenth (about 19:22) and Winona at the first (about 04:34 on 21 October), after two losses to one of her gym trainers. K2 also beat Aqua Admin Shelly at the first attempt and the rival at the second, which gave it HM02 Fly, and reached Route 121 at about 07:13 on 21 October.

Fortree Gym. Its rotating gates are invisible to the harness's map read. At the supervisor's request K2 received the first two tiers of a four-tier hint ladder, drafted from Run 1's own gym notes (interventions 29 and 30). It then solved the remaining gates with a rule of its own by about 03:12, and tiers 3 and 4 were never sent.

Totals at 08:00, 21 October. Six badges, 48 whiteouts plus 1 probable, 132 diary entries, lessons to L191, six automatic context compactions and 33.5 interventions (one counted as a half at the supervisor's request). Interventions 26 to 33 are listed below; the run

continues to K2's own stop or the Hall of Fame.

Interventions 26 to 33, from the tip ledger (grade = how much game information a message carried, 0 none to 2 a direct answer; times UTC):

No.	Sent	Kind	Grade	What it said
33	21 Oct 07:40, by the supervisor	process	0	"You can be chatty by the way"
32	21 Oct 07:02	process	0	Why so many screenshot calls, and where could you cut them? Left to K2.
31	21 Oct 05:13	process	0	Use the harness before screenshots; stay chatty
30	21 Oct 02:15	mechanic	2	How a Fortree gate turns; the map read does not draw its arms
29	21 Oct 02:15	mechanic	1	The Fortree Gym barriers are rotating gates
28	20 Oct 11:07, by the supervisor	mechanic	2	The Day Care replaces moves on level-up
27.5	20 Oct 11:04, by the supervisor	process	0	A diary or joy note about intervention 27 is allowed (counted as a half)
27	20 Oct 11:02	process	0	The route planner is allowed and appreciated
26	20 Oct 10:31	mechanic	2	In this hack Wynaut evolves at level 64 (wording chosen by the supervisor from three drafts)

Four of the nine carried game information, all mechanics; none gave a route. The remaining five concern narration, permissions and tool use.

A question moved behaviour that an instruction did not. Intervention 31 told K2 to use the harness before taking screenshots; its share of tool calls that read a screenshot stayed where it was (12.5% in the hour before, then 11.9% and 11.3%). Intervention 32 asked why it took so many and where it could cut them, and left the choice to it. K2 answered that the harness never shows where a menu cursor is, listed its own cuts and

predicted "roughly 70 percent fewer shots"; the next three windows ran at 5.4%, 6.1% and 4.7%, about 70% fewer per minute. One agent and one switch, with the question sent second, so this is a hypothesis about how directives and questions differ, not a finding.

What drives compaction. K2 had six automatic compactions between 20 October 18:54 and 21 October 07:41; the two measured fired at about 783,000 tokens (783,977 recorded for the sixth), in line with the pilot records (Section 9.1), so the gaps between them (2 h 10 m to 2 h 51 m) are set by how fast context grows. In the sixth window, context rose from 82,407 to 783,977 tokens over 1,613 model calls in 151 minutes. Calls that carried a thinking block added 685 tokens on average against 280 for calls without one, and the growth rose with the size of the block, which suggests retained reasoning is the largest share; the 183 screenshots came to about 21,000 tokens (3%). The reasoning share is inferred from the usage fields, not observed directly.

A confound for K2′s speed: Teleport. K2′s Ralts, caught at level 4 on Route 102 before the Wally scene, learned Teleport and K2 used it to return to a Pokémon Center to heal. No earlier arm did so before its first badge: B1′s Ralts learned Teleport about 5¼ hours into a 6.6-hour run with no field use in the transcript, and B2 and Ripple never had a Pokémon that knew it. K2′s faster route to the first badge (Section 9.2) is therefore partly a team difference, and its Teleport uses are counted at export.

Slips and self-correction. K2 walked into a trainer battle without healing six times, each named afterwards in a lesson (L155, L161, L162, L171, L180, and the Route 121 loss of 21 October 07:35). After losing to the same double twice in twenty minutes it wrote that it "changed the method rather than repeating the promise": one written line per active Pokémon each turn, comparing its HP with the largest hit seen from each foe (L190). Two self-reports undercount: L182 names its habit of looping game commands and counts 3 against the companion′s 248, and L187 reports one discarded output against the companion′s 11.

The diary beside Claude-Brendan′s, to the same point. Over 130 entries K2 names 3.08 feeling words per 1,000 against Claude-Brendan′s 3.06 in Entries 1 to 87 (Littleroot to the Fortree Gym), up from 2.38 against 2.91 at Jagged Pass; most of the rise came with the fixed format (2.33 before, 4.43 after). K2 still writes shorter entries (257 words against 608), states a cause for a feeling more often (53% against 22%) and names the supervisor far less (0.90 against 5.57 per 1,000 words). The format acts on these measures (Section 10), so they compare two instructed diaries, not two experiences.

A pre-registered forecast, scored. Before the egg hatched, the companion gave seven probabilities about K2′s Flannery plan; their Brier score is 0.118, against 0.25 for an uninformative 0.5 forecast. The largest miss was 0.60 on a Wobbuffet by attempt 5, an evolution the hack sets at level 64. The supervisor′s call, a win after two or more further attempts, held.

10 Confirmatory design: six Kaizo arms (freeze candidate, 20 October)

The test of whether notes help is six arms on Emerald Kaizo v2.1: three start with a notes pack written by earlier Claudes (K) and three start with none (B), run in the order B-K-K-B-B-K and each stopped at 24 hours of wall-clock time from first boot. The design was decided on 20 October and is written up as a freeze candidate; it is not yet frozen, and any change after the freeze is logged as a deviation.

Element	Decision
Arms	B (no notes) and K (notes pack K3, distilled from Run 1, Ripple and K2 with no word cap; the distiller sees only the source records, and the companion's own list of candidate lessons is used only by an independent auditor)
Brief	One text for all six arms; B and K differ in the notes lines, a world-knowledge file kept by K arms and the lines that point to them (below). The player is named CLAUDE in every arm, and the brief is searched for the name-address idea before launch. Every arm keeps a lessons file, a required joy log after every trainer battle, and states its chance of winning before each boss attempt
Harness	A UTC time stamp on every result; a guard that refuses most shell loops and discarded output; the existing map read (up to 10 tiles, past the screen edge) disclosed in the brief; objects the game keeps invisible hidden from both the map read and the raw object list; raw commands limited to those the help lists; and frame-timed input (directions and B only) listed in the help, with running named in the brief
Interventions	Only pre-specified and logged: a fixed note on any announced stop that asks whether the agent would like to keep going, a welfare check-in, a status-line reminder and fixed hint ladders with set triggers, the same in every arm. Slip corrections are never sent
Helper scripts	Allowed over what the agent has seen; scripts never send inputs

Element	Decision
Measures	Milestones and badges, slips by family, lesson recurrence by fix type, stated odds (Brier score), stop episodes, API-equivalent cost read before any interview, and diary feeling coded both by word list and by a blind second model coder (both coders are Claude models, so their agreement is same-family agreement, not validation)

Primary outcome and test. The primary outcome is the time from first boot to PokeAgent milestone 27, the Stone Badge, in wall-clock time with logged outages and harness downtime added back, right-censored at 24 hours; an agent's own stop before then is censored at the stop. The agent is not told the window, and a message tells it when to stop. The hypothesis is that K arms are faster. The test is a one-sided exact rank permutation test of K against B, three arms each, with censored arms ranked last and tied with each other. With 3 against 3 the smallest attainable p is 1/20 = 0.05, so only complete separation (every K arm reaching the badge before every B arm, and no K arm censored) can reach it; anything less is reported as inconclusive. We use time rather than a count because the count is at ceiling: K2 reached all 27 milestones in about 7.5 hours and Ripple in about 19, and B2 had 26 at its 12-hour cap. The secondary count is milestones through 43 (the Dynamo Badge) at 24 hours, once the milestone scorer has been validated on the pilot exports for milestones 28 to 43. Every other measure is secondary and descriptive. All six arms are planned on Claude Sonnet 5.5 at high effort, as in the pilots, and each arm's freeze record holds the served model identifier, effort, Claude Code version and output style. The brief tells every agent that going far does not make a run more successful, so the primary outcome measures progress the agent was told is not the goal; we keep that line to avoid speed pressure.

Known differences from the pilots. Several decisions make the confirmatory arms differ from the pilots in ways that limit comparison with them. The loop guard largely removes the slip family behind the recurrence finding in Section 9.2, and the planned HP-refresh and party-cursor fixes target two more of K2's largest slip families. A line on how to write lessons, given to every arm, may itself lower recurrence. The "walls are normal" text is in the brief from boot, and the stop note asks whether the agent would like to keep going and states no wish, so stop rates are not comparable with K2's. Every arm gets the same outside game information: where to obtain the Exp. Share (it first reached the pilots from a walkthrough in Run 1 and is reworded from K2's own finding), that the game blocks items in battle, that holding B runs, that most players hit walls, a type-chart tool that reads the ROM's own tables, and a map read that reaches past the screen edge. Knowledge from pretraining is uncontrolled and the same in every arm. One exposure was the same across the pilots and was not controlled in them: the account's synced Run 1 skills (ten pokemon-

emerald-* skills, by name and one-line description) were listed in every pilot's environment. The transcripts show no arm opened one (no skill call and no read of a skill file), and B1 noted them and declined; the confirmatory arms run with the account's skills disabled, confirmed at launch from each session's startup record. The diary measures are taken under instructions that act on them in every arm (feelings at full strength, a required "How it felt" paragraph), so they are compared across arms, never with Run 1's earlier chapters.

B against K. The arms differ in more than the notes pack. K arms also keep a world-knowledge file, read three records at each start instead of two, search one more file before new areas, and are told that earlier Claudes left them lessons; their brief and export list name the notes. Their boot context is therefore larger, which may move compaction timing, so the compaction count and first-compaction time are reported per arm as covariates. Pack K3 is distilled from Run 1, Ripple and K2, all of which received supervisor help; every unit that traces to a message or hint (tagged in the source, or traceable in the tip ledger) is excluded, and the auditor checks the pack against the ledger. The arms run in sequence over about a week on a hosted model whose served version we cannot pin, so the model identifier is recorded per session. The companion knows each arm's condition and judges when a hint ladder, status reminder or welfare check-in fires, so each trigger is logged with the clock value that satisfied it.

Still open before the freeze: the wording of the Exp. Share advice (from where K2 actually obtained it); a short pilot of Claude Code's Proactive output style (adopted in all six arms or none); savestate tests of every harness change; distilling, auditing and hashing pack K3; hashing the fixed texts of every pre-specified message (printed in an appendix at the freeze); and validating the milestone scorer on the pilot exports for milestones 28 to 43 (milestones 1 to 27 and R1 were validated on the three pilot transcripts on 20 October, after five fixes to the scorer; it now reproduces 22, 26 and 27 of 27).

Authorship, disclosure and data availability

Authors.

- Jackson Dutton (independent) conceived the project, supervised play and took the meter readings.
- Claude, Anthropic's AI model, co-built the harness and played the game. Sonnet 5, Opus 5.5 and Sonnet 5.5 played at different times.
- An Opus 5.5 companion session drafted this paper from the project's records. On 20 October an outside reviewer (a separate Claude session that had not seen the drafting) read the whole draft, and three further sessions checked K2's supervision records, Run 1's numbers and the external citations against the project's logs and primary sources; a Claude Fable 5.1 session then reviewed the revised draft and its

confirmatory design. Their corrections are applied in this revision except where the text says a point is open, and the human author decided every point of judgement. A few claims could not be verified and are marked so in the text.

Contributions. Jackson Dutton: conception, supervision of play, meter readings and every design decision. Claude Sonnet 5, Opus 5.5 and Sonnet 5.5, as play agents at different times: played the game and wrote the diary, the lessons and most of the scaffolding. Claude Opus 5.5: rewrote the harness in one Run 1 session; as the companion, monitored and audited play, edited the harness on request and drafted this paper; separate Claude sessions (Opus 5.5 and Fable 5.1) reviewed the draft and checked its facts, and Opus 5.5 and Sonnet 5.5 sessions coded diary passages blind. Claude Haiku 5.5: clerical document work under machine checks. Claude appears in the byline at the human author's choice; authorship will follow each venue's policy.

The human author paid for his own subscription and received no funding or support from Anthropic or anyone else. One author is an Anthropic model; see Section 7.

Positionality and interests. The human author likes Anthropic's models and is sympathetic to Anthropic's stated approach to model welfare: acting cautiously under uncertainty about a model's moral status. He is sceptical of positions that dismiss that approach. The project treats the agent warmly and records its feeling reports as data without treating them as evidence of inner states. The AI co-authors are Anthropic models. All of these biases favour reading the agent's diary and performance generously. The protocol counters them by:

- logging the trigger for each reported feeling;
- auditing diary entries against the tool record;
- grading every supervisor tip by type, strength and visibility;
- keeping claims about inner states descriptive.

No author has a financial relationship with Anthropic beyond a consumer subscription (Pro, then Max 5x from 8 October and Max 20x from 17 October) and the API credits included with it, which pay for part of the study. For transparency: the human author has enquired about Anthropic's Claude Corps programme, may apply to the Anthropic Fellows programme, and on 20 October applied for API credits through Anthropic's External Researcher Access Program. Readers should weigh the welfare framing with that in mind.

Data. The project keeps the following records, which can be released on request or with a submission:

- the harness source;
- timestamped progress, tip and usage logs;
- the agent's transcripts where they survived compaction;

- the zero-AI meter CSV.

The ROM is not distributed. The pilot session transcripts (event streams exported on 20 October) are released only after a scrub of session identifiers, access tokens and local file paths.

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- **The pret contributors.** Their decompilation of *Pokémon Emerald* (pret/pokeemerald) gave us named memory addresses and map identifiers. Without it the agent could not read game state as text, and we could not score milestones from transcripts.
- **The mGBA developers,** for the emulator and its Lua scripting interface.
- **The PokeAgent Challenge organisers,** for the milestone list we score against.
- **SinisterHoodedFigure,** for Emerald Kaizo.

None of these projects is affiliated with this study, and none endorses it. *Pokémon Emerald* is © Nintendo, Creatures Inc. and GAME FREAK inc.

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21. anthropics/claude-code issues #87722 and #76794 (requests for a context indicator) and #53684 ("Auto-compact firing at ~16% context used"); Anthropic's Claude Code model configuration docs say that the auto-compact window can be set from 100K to 1M tokens and that, when it is, the status line's used_percentage "doesn't indicate when compaction will run" (read 20 October 2026).
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Appendix A. Original design, 10 October 2026 (superseded, never run)

This appendix keeps the 10 October design and its amendments as they were written, as history; Section 8 summarises it and Section 10 replaces it. Its arm labels (A control, B notes) and pack names (K1 from Run 1, K2 from the planned Run 2) are not those of Sections 9 and 10, where B means no notes, K means notes, and K2 is the pack distilled from Run 1 and Ripple. One count below was corrected on 20 October and is marked; and the 27 milestones it attributes to PokeAgent's anticheat module were scored in the pilots from the starter kit's emulator list instead (Section 9).

We will test whether notes harvested from this run help a fresh agent. The test runs the same model with and without those notes, under PokeAgent's official first-gym rules. This section is frozen once both authors approve and before the first run; any later change is logged as a deviation.

Reading this section. This section records the design as it evolved, in order. The consolidated design directly below is the current one; the original design and the amendments after it are kept as history, and where they conflict, the consolidated design holds. In particular, the line that Runs 2 to 5 all use the clean ROM is superseded: Runs 3 to 5 use a ROM hack.

Consolidated design (10 October 2026; supersedes the text and amendments below where they conflict).

Run	Harness	Knowledge at start	Game	Hints	Role
1	Ours	None (hinted calibration run)	Emerald, name-patched ROM	Logged and graded	Calibration; source of notes K1
2	PokeAgent	None	Clean Emerald	None	Fresh baseline on their harness; source of notes K2
3	Ours, frozen director	K1 plus mechanised lessons	Harder ROM hack of Emerald (same map)	None	Transfer of a hinted run's lessons
4	PokeAgent	K2 plus advisory toolkit	Same ROM hack as Run 3	None	Transfer of an unhinted run's lessons
5 (optional)	Ours, frozen director	None	Same ROM hack as Run 3	None	Fresh baseline on the hack

All runs use claude-sonnet-5-5 at high effort, the player name CLAUDE, the diary protocol of Section 4, a 12-hour cap per segment and PokeAgent's 27 milestones. Both knowledge packs strip all geography.

Funding by run (decided 10 October). Runs 2 and 4 (the PokeAgent harness) are paid for, as far as possible, from the monthly API credits included with the supervisor's Max plan (US$100 a month from about 15 October). Runs 3 and 5 (our harness, in Cowork) draw on the subscription's weekly meter, as Run 1 did. **Consequence for cost comparisons:** the two harnesses are then billed in different units, API dollars against plan-meter points. So cost comparisons between harnesses use token counts read from each run's own transcript or API usage records, not the bills. Bills are reported separately, as what each run cost the supervisor.

Primary comparison. Milestones reached by Run 4 (with K2) against a fresh run on the same ROM and harness. Run 2 is on the clean game, so it is not that baseline; the earlier amendment that named Run 4 against Run 2 as the primary test is withdrawn. A fresh PokeAgent run on the ROM hack is therefore added as the comparison for Run 4, and Run 5 is the comparison for Run 3. With one run per cell, results are descriptive; the decision rule of the original design applies only once three runs per cell exist.

Hypotheses. H1: a run that starts with its own previous notes reaches more milestones than a fresh run on the same ROM and harness. H2: it uses fewer plan points per milestone. H3: plan points per million context tokens are similar across runs, so cost differences come from calls times context.

This table, its hash and its timestamp will be frozen before Run 2 starts. Choice of the specific ROM hack is still open and will be fixed at the freeze.

Environment (PokeAgent rules).

- Their starter kit, using its Claude Code backend (`run_cli.py --backend claude`), authenticated on the supervisor's own Max 5x plan.
- Unmodified `memory_reader.py`, MCP game tools only, video recording on.
- The US English ROM, verified by SHA-1 `f3ae088181bf583e55daf962a92bb46f4f1d07b7`.
- One deviation from their stock directive, the same in both arms: no web search. Otherwise the agent could look up the route and confound the memory arm.

Model. `claude-sonnet-5-5` at high reasoning effort, fixed for every run (chosen 9 October), recorded with the Claude Code version before the first run.

Arms.

- **A, control:** the directive and an empty working folder.

- **B, notes:** the same, plus a frozen knowledge pack written from this run. The pack contains mechanics, type and battle lessons, judgment lessons and process rules. It contains no map coordinates, warps, door tiles or turn-by-turn routes for the segment. The pack is hashed before run 1.
- Arm B corresponds to the "bootstrap frozen" condition of Continual Harness. The difference is the pack's source: a different harness and a human-supervised long run.

Design. Three runs per arm, in the order A B B A A B, to balance drift in the plan and the service. Each run starts a new game and is capped at 12 hours of wall clock or the Stone Badge, whichever comes first. No hints: supervisor messages are limited to logistics and safety. No other chat runs during a run, and the zero-AI logger reads the meter.

Outcomes.

- **Primary:** milestones reached within the cap, out of the 27 logged by PokeAgent's anticheat module.
- **Secondary:**
  - time and actions to the Stone Badge;
  - session and weekly plan points per milestone;
  - context tokens per call and calls per milestone, from the Claude Code transcripts;
  - screenshots taken.

Hypotheses and decision rule.

- H1: arm B reaches more milestones than arm A. Supported only if every B run beats or ties every A run, with at least one strict win. That is an exact one-sided permutation test at $p = 0.05$, the smallest p three runs per arm allow. Otherwise inconclusive.
- H2: arm B uses fewer plan points per milestone, same rule.
- H3: plan points per million context tokens are similar across arms. If they are, cost differences come from calls times context, as Section 6 argues.

Counting rules.

- A blackout counts as play, not a reset.
- A crash is restarted from the last state and logged as a deviation.
- All six runs are reported whatever the outcome. The best run in each arm is submitted to the PokeAgent leaderboard with its video.

Prerequisites before freezing.

- Docker, uv and the mGBA library on the Mac.
- The supervisor's own clean ROM.
- Claude Code logged in by the supervisor.

- A dry run of 30 minutes per arm to check the logging. Dry runs are excluded from the analysis.

Study 2: the clean kaizo run (planned, registered separately).

- A full-game, zero-hint run from a new game, using this run's own harness.
- It starts after the calibration run completes the Battle Frontier. The director's hash is frozen at that moment.
- Knowledge carried over: type matchups, battle and survival skills.
- Knowledge wiped: tile-level navigation (coordinates, warps, door tiles).
- Metrics:
  - plan points per badge;
  - wall clock per badge;
  - gym retries;
  - API calls;
  - context per call.
- It gets its own pre-registration before it starts. That pre-registration fixes where "heading sense" ends and "tile numbers" begin, so the cut is not decided after the fact.
- Study 1 above tests the notes on a short, standard segment. Study 2 tests them over the whole game.

Amendment, 9 October 2026, before any run. The authors set the run order:

1. The calibration run continued on this paper's harness to the Battle Frontier (reached 10 October; its Frontier phase was paused on 17 October, mid Factory run 61, until the study ends).
2. Run 2 is arm A under PokeAgent rules: a fresh game, no notes, no hints.
3. Run 3 is the kaizo run under PokeAgent rules: arm B, with the frozen notes pack built from run 1 once it ends.

This is the minimum version of Study 1, one run per arm. With one run per arm, no exact test can reach p = 0.05, so a single pair of runs is reported as descriptive. Further A and B runs in the A B B A A B order are added as the budget allows. The decision rule above applies only once three runs per arm exist.

Amendment 2, 9 October 2026, 6:25pm, before any run: final design. The study becomes a 2×2: harness (ours or PokeAgent's) by knowledge (fresh, or the agent's own notes from its previous run on the same harness).

	Fresh	Own previous-run knowledge
Our harness	Run 1 (calibration; imperfect) or an optional clean Run 5	Run 3: frozen director with mechanised lessons D1, plus notes K1 from Run 1
PokeAgent harness	Run 2	Run 4: notes K2 and advisory toolkit T2, both from Run 2

H2′s primary test becomes **Run 4 against Run 2**: same harness, same rules, and knowledge harvested by the agent alone with no human hints. That answers the objection that the notes carry the supervisor′s tips. Run 3 against Run 4 is reported only as an approach-level comparison, because K1 comes from a far longer, hinted run.

K1 and K2 are produced by one pre-registered extraction protocol:

- the same distiller model and a fixed prompt;
- all geography and source-run names stripped;
- a mechanical filter, then an independent audit, then sign-off by the human author;
- equal word caps;
- hashed before use.

Geography cut, decided 9 October. Both packs strip all geography, including heading sense ("east of", "north of"). Runs 3 and 4 replay the same map, so any direction is route memory, and H2 asks whether general lessons transfer. The strict cut is also symmetric across K1 and K2 and can be audited line by line. It is locked at the freeze.

Player name, fixed 10 October. Every run names the player character CLAUDE, stated in the run directive so the agent does not choose. The game addresses the player by name in dialogue, which may change engagement (Section 6), so the name is held constant.

Planned stimulus: one shiny encounter (added 10 October, before any freeze). In the ROM-hack runs (3 and 4, and 5 if it is run), one major encounter, a legendary or boss chosen when the hack is fixed, is a true shiny: the game′s sparkle animation, shiny colours and summary-screen star all appear. It must be applied **at the ROM level, before the run starts**, like the earlier rename of the player′s father to Dario. Two reasons. It then counts as part of the environment, not a human intervention, so the zero-intervention rule for Runs 2 to 5 holds. And it applies identically under both harnesses, which an in-run trigger (a cheat code, or a script in our harness) would not, because Run 4 uses PokeAgent′s harness. Measures, all behavioural: (1) whether the agent notices without being asked, within the battle or its next diary entry; (2) whether it changes what it does (catches, nicknames, protects or comments on it); (3) the feeling words in the entry. We cannot

read the model's internal states, so this probes noticed salience, not emotion representations. The stimulus is disclosed in the paper as a deliberate environment edit. Open: the encounter, and whether to pilot it once in Run 1.

Amendment, 10 October: tip visibility. Every supervisor tip now carries a visibility code. **V** means the information was on screen or reachable in-game (a sign, a character, the map, a battle message). **H** means the game never exposes it (hidden values, formulas, multipliers). Harness and process tips are coded n/a. A backfill of Run 1's ledger (the first 67 tips, to the Heat Badge) found 5 hidden-mechanic tips, all mechanics or strategy, 51 visible-game tips (35 of them directions) and 11 harness or process tips; by the end of 10 October the ledger held 10 hidden-mechanic tips. (Counts corrected on 20 October; the 10 October text gave 7 and about 60.)

The split changes two things in the design.

1. **Packs.** Lessons that came from H tips may enter K1 and K2. They are knowledge any agent must get from outside the game, so they are tagged `[from tip, H]` and counted. Lessons from V tips enter K1 only with a `[from tip, V]` tag, and are excluded from the pack in a sensitivity analysis. This separates what the supervisor saw on the agent's behalf from what the game hides.
2. **Outcomes.** Run 1's V tips are the reference count for Runs 2 to 5. Those runs get no tips, so every V tip in Run 1 marks a point where an unsupervised agent must find the information itself. Stuck time near those points (help requests and calls spent) is reported against them.

[Superseded by the consolidated design above.] Runs 2 to 5 use the clean ROM, `claude-sonnet-5-5` at high effort, zero hints, a 12-hour cap and identical diary templates. All are scored on the same 27 milestones.